

Practice Questions for 2024-11-19-MaxFlowMinCut

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1 Questions

1. In the context of the Max Flow Min Cut theorem (Slides 1-4), if we have a network flow problem where the maximum flow from the source to the sink is 10 units, what can we definitively conclude about the minimum capacity of any cut separating the source and the sink?
 - a. The minimum cut capacity is less than 10 units.
 - b. The minimum cut capacity is greater than 10 units.
 - c. The minimum cut capacity is exactly 10 units.
 - d. There is no relationship between the maximum flow and the minimum cut capacity.
 - e. The minimum cut capacity can be any value greater than or equal to 10 units.
2. Consider the network flow problem illustrated in Slide 5. Suppose we find a cut (C, C^c) such that the capacity of the cut, $K(C)$, is 8. What can we conclude about the maximum flow from San Francisco (SF) to New York City (NYC)?
 - a. The maximum flow is exactly 8.
 - b. The maximum flow is at least 8.
 - c. The maximum flow is at most 8.
 - d. The maximum flow is less than 8.
 - e. We cannot determine the maximum flow from this information alone.
3. The Ford-Fulkerson algorithm (Slides 10-13) iteratively increases the flow in a network. What condition must be met to guarantee that the algorithm terminates and finds the maximum flow?
 - a. The algorithm always terminates, regardless of the network structure.
 - b. The algorithm terminates if the capacities are integers.
 - c. The algorithm terminates if there are no cycles in the network.
 - d. The algorithm terminates if the network is acyclic and the capacities are integers.
 - e. The algorithm terminates if there is no augmenting path from source to sink.
4. In the linear programming formulation of the Max Flow Min Cut problem (Slide 15), the introduction of a fictitious arc from the sink to the source with infinite capacity and a cost of -1 serves what crucial purpose?

- a. It simplifies the constraint matrix.
 - b. It ensures the feasibility of the linear program.
 - c. It allows the maximum flow to be expressed as the optimal value of a linear objective function.
 - d. It prevents cycling in the simplex method.
 - e. It guarantees the uniqueness of the optimal solution.
5. In the context of the Max Flow Min Cut theorem (Slides 1-4), if we have a network flow problem where the maximum flow from the source to the sink is 10 units, what can we definitively conclude about the minimum capacity of any cut separating the source and the sink?
- a. The minimum capacity of any cut is less than or equal to 10 units.
 - b. The minimum capacity of any cut is greater than or equal to 10 units.
 - c. The minimum capacity of any cut is exactly 10 units.
 - d. There is no definitive conclusion about the minimum cut capacity.
 - e. The minimum capacity of any cut is less than 10 units.
6. Consider the network flow problem in Slide 5. Suppose we find a cut (C, C^c) with capacity $K(C) = 12$. What is the maximum possible flow from San Francisco (SF) to New York City (NYC)?
- a. The maximum flow is at least 12.
 - b. The maximum flow is exactly 12.
 - c. The maximum flow is at most 12.
 - d. The maximum flow is less than 12.
 - e. We cannot determine the maximum flow from this information alone.
7. In the context of the Max Flow Min Cut theorem (Slides 1-4), consider a network flow problem where the maximum flow from the source to the sink is known to be 15 units. A cut (C, C^c) is identified, and the capacity of this cut, $K(C)$, is calculated to be 12 units. What can you definitively conclude about the relationship between the maximum flow and the minimum cut capacity in this specific instance, and why?
- a. The maximum flow is equal to the minimum cut capacity.
 - b. The maximum flow is greater than the minimum cut capacity.
 - c. The maximum flow is less than the minimum cut capacity.
 - d. The relationship between maximum flow and minimum cut capacity cannot be determined from the given information.
 - e. The maximum flow is equal to the capacity of the identified cut (C, C^c) .

2 Answers

1. In the context of the Max Flow Min Cut theorem (Slides 1-4), if we have a network flow problem where the maximum flow from the source to the sink is 10 units, what can we definitively conclude about the minimum capacity of any cut separating the source and the sink?
 - a. This is incorrect because the Max Flow Min Cut theorem establishes an equality between the maximum flow and the minimum cut capacity.
 - b. This is incorrect because the Max Flow Min Cut theorem establishes an equality between the maximum flow and the minimum cut capacity.
 - c. The Max Flow Min Cut theorem states that the maximum flow in a network is equal to the minimum capacity of any cut separating the source and the sink. Therefore, if the maximum flow is 10 units, the minimum cut capacity must also be 10 units.
 - d. This is incorrect because the Max Flow Min Cut theorem directly relates the maximum flow and the minimum cut capacity.
 - e. This is incorrect because the Max Flow Min Cut theorem establishes an equality between the maximum flow and the minimum cut capacity.
2. Consider the network flow problem illustrated in Slide 5. Suppose we find a cut (C, C^c) such that the capacity of the cut, $K(C)$, is 8. What can we conclude about the maximum flow from San Francisco (SF) to New York City (NYC)?
 - a. This would only be true if we also knew that this cut was the minimum cut.
 - b. While possible, it's not guaranteed. A smaller minimum cut could exist.
 - c. The Max Flow Min Cut theorem states that the maximum flow is less than or equal to the capacity of any cut. Since we found a cut with capacity 8, the maximum flow cannot exceed 8.
 - d. This is too strong a conclusion; the maximum flow could be 8.
 - e. The Max Flow Min Cut theorem provides a relationship between maximum flow and minimum cut capacity.
3. The Ford-Fulkerson algorithm (Slides 10-13) iteratively increases the flow in a network. What condition must be met to guarantee that the algorithm terminates and finds the maximum flow?
 - a. The algorithm may not terminate if the capacities are not integers, leading to an infinite loop.
 - b. While integer capacities help ensure termination, it's not a sufficient condition.
 - c. Cycles do not prevent termination; the algorithm handles them.
 - d. Acyclicity is not a necessary condition for termination.
 - e. The Ford-Fulkerson algorithm terminates when no augmenting path (a path from source to sink with available capacity) can be found. This indicates that the maximum flow has been reached.
4. In the linear programming formulation of the Max Flow Min Cut problem (Slide 15), the introduction of a fictitious arc from the sink to the source with infinite capacity and a cost of -1 serves what crucial purpose?
 - a. While it might have a secondary effect on the matrix, this is not its primary purpose.
 - b. Feasibility is ensured by the other constraints in the formulation.

- c. The fictitious arc transforms the problem of maximizing flow into a minimization problem where the objective function directly represents the maximum flow. Minimizing the negative flow on this arc is equivalent to maximizing the flow in the network.
 - d. Cycling is prevented by other mechanisms in the simplex method.
 - e. Multiple optimal solutions are possible in network flow problems.
5. In the context of the Max Flow Min Cut theorem (Slides 1-4), if we have a network flow problem where the maximum flow from the source to the sink is 10 units, what can we definitively conclude about the minimum capacity of any cut separating the source and the sink?
- a. This is incorrect because the minimum cut capacity cannot be smaller than the maximum flow.
 - b. This is incorrect because the minimum cut capacity cannot be larger than the maximum flow.
 - c. The Max Flow Min Cut theorem states that the maximum flow in a network is equal to the minimum capacity of any cut separating the source and the sink. Therefore, if the maximum flow is 10 units, the minimum cut capacity must also be 10 units.
 - d. This is incorrect because the Max Flow Min Cut theorem provides a direct relationship between maximum flow and minimum cut capacity.
 - e. This is incorrect because the minimum cut capacity cannot be smaller than the maximum flow.
6. Consider the network flow problem in Slide 5. Suppose we find a cut $(C, C^{\sup} < c < /sup>)$ with capacity $K(C) = 12$. What is the maximum possible flow from San Francisco (SF) to New York City (NYC)?
- a. The algorithm terminates when no augmenting path (a path from source to sink with available capacity) can be found. At this point, the maximum flow has been achieved.
 - b. Incorrect. The maximum flow is equal to the minimum cut capacity. We have found a cut with capacity 12, but there might be a smaller cut capacity.
 - c. The Max Flow Min Cut theorem states that the maximum flow is equal to the minimum cut capacity. Since we found a cut with capacity 12, the maximum flow cannot exceed 12.
 - d. Incorrect. The maximum flow is equal to the minimum cut capacity. We have found a cut with capacity 12, which is an upper bound on the maximum flow, but it might be equal to 12.
 - e. Incorrect. The Max Flow Min Cut theorem directly relates the maximum flow to the minimum cut capacity. The capacity of the cut provides an upper bound on the maximum flow.
7. In the context of the Max Flow Min Cut theorem (Slides 1-4), consider a network flow problem where the maximum flow from the source to the sink is known to be 15 units. A cut $(C, C^{\sup} < c < /sup>)$ is identified, and the capacity of this cut, $K(C)$, is calculated to be 12 units. What can you definitively conclude about the relationship between the maximum flow and the minimum cut capacity in this specific instance, and why?
- a. The Max Flow Min Cut theorem states that the maximum flow is equal to the minimum cut capacity. However, this question provides a specific example where a cut with capacity 12 is found, while the maximum flow is 15. The theorem holds true for the minimum cut capacity, not necessarily for any arbitrary cut.
 - b. The maximum flow (15 units) is greater than the capacity of the identified cut (12 units). This is consistent with the Max Flow Min Cut theorem, which states that the maximum flow is equal to

the *minimum* cut capacity. The identified cut is not necessarily the minimum cut; there might be other cuts with smaller capacities. The theorem guarantees that the minimum cut capacity will be equal to the maximum flow, but it does not imply that every cut will have a capacity equal to the maximum flow.

- c. The Max Flow Min Cut theorem states that the maximum flow is equal to the minimum cut capacity. Finding a cut with capacity 12 does not imply that the minimum cut capacity is 12; there might be other cuts with smaller capacities. The maximum flow (15) can be greater than the capacity of a particular cut, but it will always be equal to the capacity of the minimum cut.
- d. The Max Flow Min Cut theorem provides a direct relationship between the maximum flow and the minimum cut capacity. While the capacity of a specific cut is given, the theorem allows us to conclude something about the relationship between the maximum flow and the minimum cut capacity.
- e. The maximum flow is 15, while the capacity of the identified cut is 12. The theorem states that the maximum flow is equal to the minimum cut capacity, not necessarily the capacity of any arbitrary cut.